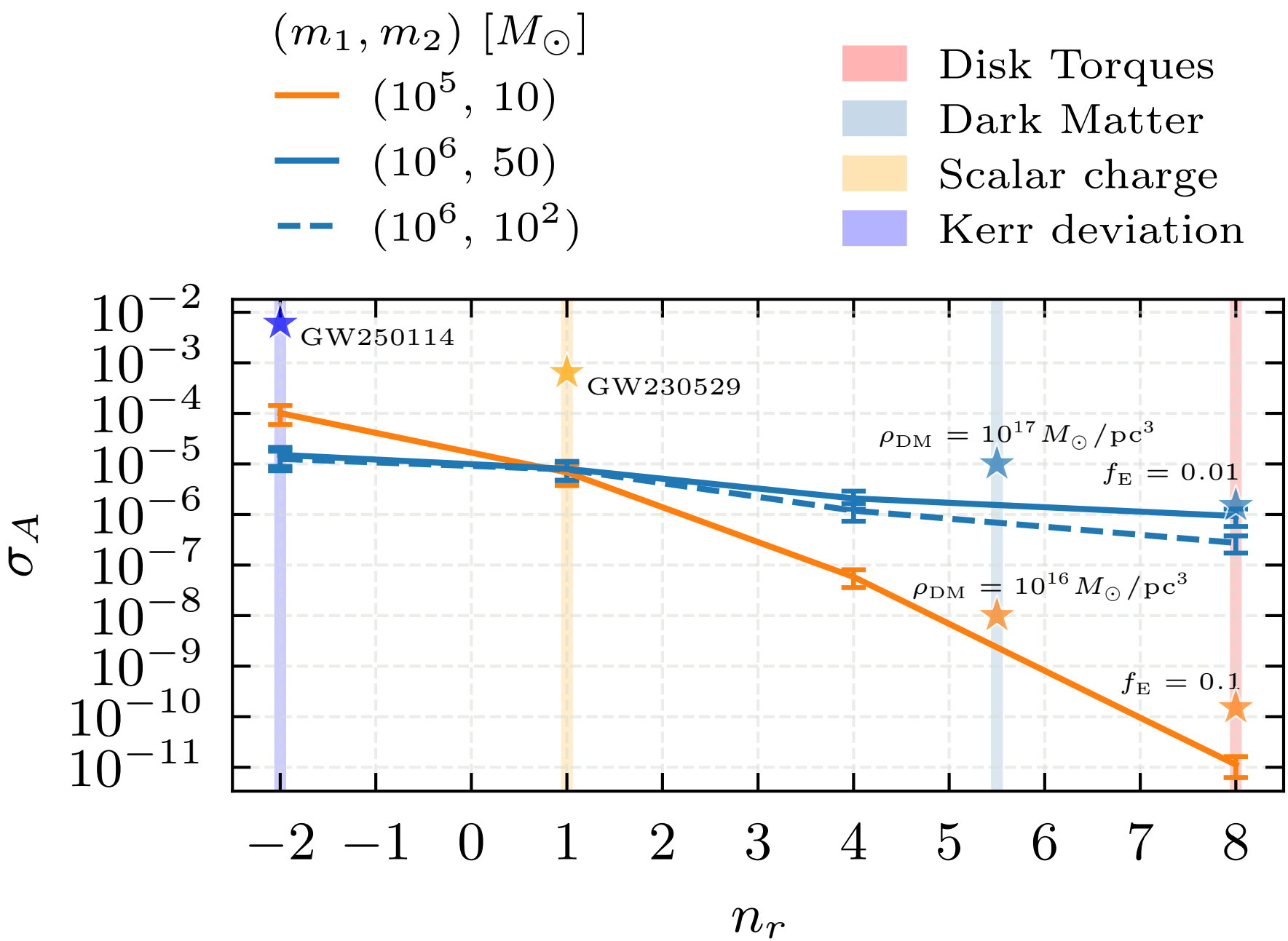
(m_1, m_2) $[M_\odot]$
$(10^5, 10)$
$(10^6, 50)$
$(10^6, 10^2)$
Disk Torques
Dark Matter
Scalar charge
Kerr deviation
GW250114
GW230529
$\rho_{\rm DM} = 10^{17}\, M_\odot/{\rm pc}^3$
$f_{\rm E} = 0.01$
$\rho_{\rm DM} = 10^{16}\, M_\odot/{\rm pc}^3$
$f_{\rm E} = 0.1$
σ_A
n_r
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
10^{-11}
-2 -1 0 1 2 3 4 5 6 7 8